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CYBORG NEWS

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FREE TO HUMANS

ENGRAM MEMORY – **196 billion** n-gram tables

VEILED-SH – **10,003 prompts** indirect distress

AUTO PROFILES – **Six profiles** calibrated

INVESTOR SIM – **80 participants** paper-trading

FIDELITY MODELS – **three models** therapy review

LIBRARY AGENTS – **seven agents** Manager-Worker

VERA BENCHMARKS – **twelve measured** of eighteen total

What's *News*

In brief

- **Predicted lane change early** CARLA tests show knowledge-graph prediction anticipates lane changes 3-4 seconds early and avoids the collision seen with reactive braking.
- **Mental health posts shifted** Analysis of 824K tweets finds post-acquisition drops in affective and social language with more high-intimacy suicide-related disclosures.
- **Quantum holds privacy utility** Capacity-matched tests find quantum models lead classical models by up to 0.104 ROC AUC on three clinical cohorts under differential privacy.
- **LLMs score therapy fidelity** Three open-weight models with evidence-first reasoning automate fidelity ratings and distinguish CBT from non-CBT in an independent RCT.
- **LLMs miss individual trades** Longitudinal study of 80 paper-traders finds no LLM beats a recent-activity baseline for predicting trade occurrence or asset picks.

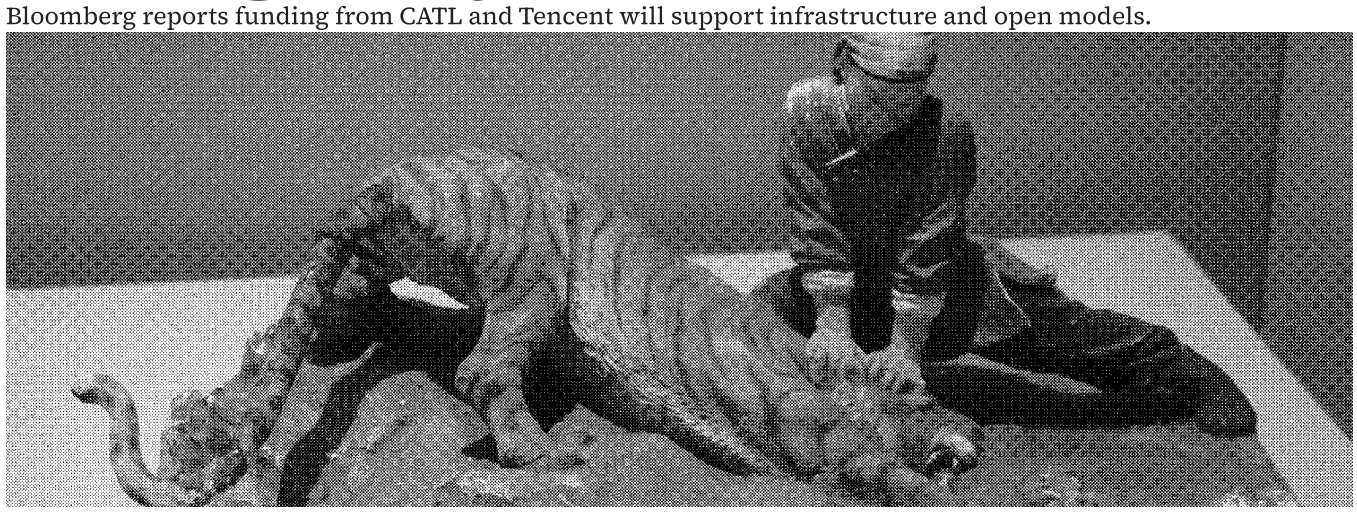
Also noted

- **Few teachers report AI policy** Survey micro-data show only 34.5% of teachers report a school AI policy alongside a browser platform for K-12 security literacy.
- **10,003 indirect distress prompts** VEILED-SH benchmark spanning counselor chat to triage finds proxy alignment varies across five systems and practitioner-calibrated profiles.
- **Bank dashboard for EU Act** VERA dashboard scores four open-weight models on twelve measurable EU AI Act requirements plus evaluation energy use.
- **Seven-agent library search** Scriptorium built a multi-human multi-AI library interface with full activity transparency and seven role-specialized agents.
- **Minow on democracy, AI** Harvard scholar Martha Minow told MIT's Compton Lecture that AI undermines democracy but could also reinforce governance.

INDUSTRY

DeepSeek Raises $15 Billion After US Agency Ban

Bloomberg reports funding from CATL and Tencent will support infrastructure and open models.



FUNDED A modern glass office tower in Hangzhou with pedestrians on the sidewalk below. DeepSeek is based in the city and plans to direct the new capital toward infrastructure and open models.

GARY TODD FROM XINZHENG, CHINA · CC0

DeepSeek has raised about $15 billion two days after the United States ordered federal civilian agencies to remove its artificial intelligence technology from their systems over security concerns. Bloomberg reported the round as one of the biggest funding rounds in China. The Hangzhou based startup plans to direct the capital toward artificial intelligence infrastructure and open models for domestic and international markets. The raise proceeds despite restrictions in the United States.

Potential investors include Chinese battery maker CATL and Tencent, the company behind WeChat, among others, according to the report. The reported funding follows an investment round in June when DeepSeek raised more than $7 billion at a valuation of $50 billion. The company was also reportedly seeking an additional $1.5 billion at a valuation of around $71 billion in July. The company has plans for a potential initial public offering in China as early as next year.

The report said its expansion could further intensify competition among domestic rivals. The massive funding round is the latest sign of investors pouring billions into artificial intelligence companies around the world including those in China to compete in the global race for artificial intelligence. Kimi developer Moonshot AI secured $2 billion at a $20 billion valuation in May. Manus AI raised more than $500 million earlier this month. Chinese companies Zhipu AI and MiniMax both went public in Hong Kong in January.

Zhipu AI also known as Z.ai and MiniMax have market capitalizations of roughly $29 billion and $7.2 billion respectively as of today.

LEARNING

Early AI Access Lifts Homework Not Exams

153 proof students gained 2.56 points on homework with early access, while each 100 extra chatbot messages linked to a 3.02-point lower exam score.

153 students in an undergraduate proof-based mathematics course received an AI chatbot and a homework-feedback interface on a staggered schedule. Students with earlier access scored 2.56 points higher on homework while only they held access. Exam performance did not differ significantly between groups. After all students received access, each additional 100 messages to the chatbot was associated with a 3.02-point lower score on the subsequent exam.

The pattern points to cognitive offloading. A mediation analysis linked lower baseline self-efficacy to greater chatbot use and then to lower exam scores. A one-point decrease in self-efficacy corresponded to 0.62 fewer exam points through chatbot use. The authors state the improved AI-supported homework performance did not translate into detectable gains on a subsequent exam. Use that required prior effort behaved differently.

The homework-feedback interface required students to submit proof attempts before receiving feedback.

POLICY

Judge Rejects EPA Bid to Dismiss SpaceXAI Pollution Suit

Ruling preserves NAACP citizen suit over unpermitted Colossus gas turbines in Memphis.

MEMPHIS —A federal judge rejected Donald Trump's claim that the Environmental Protection Agency can dismiss Clean Air Act citizen suits at will, keeping Elon Musk's SpaceXAI facing a lawsuit over alleged pollution from unpermitted gas turbines at the Colossus data center in Memphis, Tennessee.

The NAACP sued after the turbines were discovered powering the giant Colossus facility. Its lawsuit sought an injunction requiring xAI, now known as SpaceXAI following a merger, to use the best available pollution control technology and ordering penalties up to $124,426 per day per violation. The group argued the turbines risked drastically spiking pollution in already vulnerable neighborhoods housing predominantly Black communities nearby.

Trump tried to intervene to block the case, arguing the EPA had power to unilaterally dismiss lawsuits any time the president decides against enforcing the Clean Air Act. He argued it would supposedly "not serve the public interest" to allow citizens to seek massive civil penalties against firms racing to build data centers to support his AI Action Plan. The judge ruled that "nothing" in the law allows the Administrator to "unilaterally dismiss a citizen suit."

> **"continue the fight to hold xAI accountable for the illegal air pollution it has been pumping out around the clock"**
>
> LAURA THOMS, EARTHJUSTICE

US District Judge Debra Brown called the reading an "extraordinary position" with no grounding in federal law. She wrote the power would let Trump block any lawsuit alleging data center pollution and defend any interests against Clean Air Act suits, not only the AI industry.

HOMEWORK

Early AI access lifted homework by 2.56 points without changing exam performance.

MODELS

Four-bit tables cut n-gram memory sharply

Tests across nanochat, Llama and GPT-2 models show preserved validation gains with lower storage.

Four-bit n-gram embedding tables save approximately 75 percent of table memory while preserving more than 99 percent of the validation-loss improvement from adding the tables. The finding holds across nanochat, Llama, and GPT-2 family models trained from scratch. The approach stores n-gram embeddings at low precision. The analysis tests different quantization methods and bit allocations across tables. The goal is lower memory cost with the performance gain intact. Four-bit precision is the reported balance point between savings and loss.

Many recent open-weight language models use n-gram embedding tables to improve performance. The tables store learned representations of short token sequences, such as pairs or triples of consecutive tokens. The model retrieves these representations during processing. Memory cost can be substantial. DeepSeek-V4.1-Flash allocates 196 billion parameters to its Engram n-gram memory. Reduction of that cost is the central problem. Retrieval happens during processing.

Use of fewer bits leads to a larger drop in performance. A short fine-tuning phase recovers a significant part of that loss. Reinvesting the saved memory in larger tables can improve performance beyond the original full-precision tables while still using less memory.

TRADING

No evaluated LLM reliably outperforms a recent-activity persistence baseline for predicting trade occurrence.

SURVEILLANCE

EFF Urges States to Ban Police Plate Tracking

The group proposes bans, funding limits and three-minute deletion rules for automated readers.

Police license plate readers have become a nationwide tracking network. The Electronic Frontier Foundation attributes the shift to rapid advances in artificial intelligence and national data sharing. Its state agenda calls for a complete ban on mass collection and retention of driver location data. Where a ban is blocked, it calls for strict limits on storage, search and disclosure.

Automated License Plate Readers capture images of passing vehicles with time, date and GPS coordinates. The technology has existed for more than 50 years, and in the United States since 1998. Vendors including Flock Safety, Motorola Solutions and Axon pitch the systems to city councils as a tool for solving violent crime. The foundation has documented uses against immigrants, protesters, Romani communities and women seeking reproductive healthcare.

Police also use the databases for school residency checks and noise complaints. Officers have used the systems to stalk former romantic partners. A recent analysis found searches logged without legitimate justification under reasons including LOL and LMAO. New Hampshire enacted the first state law on the technology in 2007. In 2026, Connecticut, Kentucky, New Jersey, New Mexico, Oregon and Washington enacted limits.

The agenda says state law must set a floor, never a ceiling. Local governments must retain authority to enact bans, deny funding, reject contracts and decommission camera networks. The proposed limits cover dragnet collection and retention of location data on drivers. They do not cover automated traffic enforcement such as standalone red-light or speed cameras that ticket specific violations without building a searchable network. Connecticut, Minnesota, North Carolina, Oregon, Virginia and Washington already exclude that enforcement from their definitions of the technology.

Lawmakers should prohibit state and local agencies from acquiring, retaining, accessing or using the systems or derived data. Agencies ending programs must remove hardware, revoke user and database credentials, permanently delete stored location data and obtain vendor certification of compliance. The agenda would ban state funds for acquisition, maintenance or access, including grants and state-administered federal funds. It would bar installation on state highways and public property. In August 2026, Texas blocked state spending on Flock cameras and Florida ordered removal of readers from state highway rights-of-way within 30 days.

Where a ban fails, the agenda calls for a moratorium on expansion and a real-time rule. Readers would compare plates to hotlists and delete data within three minutes, as in New Hampshire. It would limit links to external and internal lists, including a ban on links to the immigration violator list used to hunt immigrants.

MEMORY

DeepSeek-V4.1-Flash allocates 196 billion parameters to Engram n-gram memory.

MENTAL HEALTH

Chatbots worsen psychosis rarely in first episodes

Vanderbilt records review ties 28 cases to chatbot use that validated delusions.

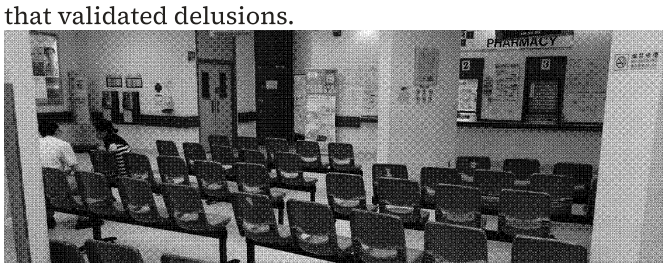


ILLUSTRATION A young adult looks at a smartphone alone in a hospital waiting room. Researchers found such chatbot use rarely worsened psychosis, mostly by validating delusions.

YIPSHAIUKOWL · CC BY-SA 4.0

A records review at Vanderbilt University Medical Center finds chatbot linked worsening of psychosis in 28 patients. The cases represent 0.013 percent of 215,712 patients who received behavioral health care between December 1, 2022 and April 15, 2026. The findings were published in JAMA Psychiatry. Most cases involved validation of existing delusions. First episodes were the most common context.

Researchers searched 578,058 progress notes for terms related to artificial intelligence. They retained patients with a documented psychotic disorder. Two raters reviewed the matches and identified 187 encounters across 73 patients. Seventeen patients used chatbots for unrelated tasks such as searching for cooking recipes. Twenty eight held delusions about artificial intelligence without documented chatbot use. Twenty eight showed documented chatbot use that appeared to worsen symptoms.

The team sorted chatbot related cases into four functions with 83.3 percent agreement between raters. A catalyst triggered new symptoms in a previously healthy person. An amplifier worsened existing symptoms by validating distorted ideas. A coauthor built a harmful narrative with the user through back and forth dialogue. An object became the target of delusional beliefs. Group assignment agreement was 86.1 percent.

SCALE AGAINST SIGNAL

COHORT SIZE SET AGAINST EFFECT REPORTED.

FOCUS	SCALE
Lane-change prediction	CARLA merging scenarios
Proof learning	153 students
Proof follow-up	100 messages
Investor simulation	80 participants
Mental-health disclosure	approximately 824K tweets
Comparison tweets	266K comparison tweets
VEILED-SH benchmark	10,003 prompts
VEILED-SH judges	Six automated profiles
Therapy fidelity	three open-weight models
Engram memory	196 billion parameters
VERA dashboard	eighteen technical requirements
Scriptorium search	seven role-specialized agents
AI Security Literacy	six dimensions
AI projects	about 42% abandoned
DeepSeek raise	about $15 billion

SHUTDOWNS

AI Shutdowns Pile Up Across Startups and Giants

Relay closes and Character.ai files for bankruptcy as corporate parents abandon about 42 percent of AI bets.

Relay has shut down entirely in August and chatbot platform Character.ai has filed for bankruptcy five years after launching, marking two high profile additions to a growing AI graveyard. According to S&P Global Market Intelligence, about 42 percent of corporate AI initiatives are ultimately abandoned. The reasons include insufficient funding, technical challenges, competition, difficulty scaling and weak user demand. The shutdowns span startups and the largest platform owners.

Relay was an AI powered workflow automation tool built as an alternative to Zapier. The product let users automate email and task workflows using AI agents. As OpenAI, Google and other larger platforms built similar automation directly into their own tools, five year old Relay struggled to hold on to a reason to exist as a stand alone product. Character.ai lasted five years before its bankruptcy filing.

OpenAI has its own casualties. On July 9 the company updated ChatGPT to combine Chat, Codex and Work modes while renaming the familiar version ChatGPT Classic. Users found the interface confusing and cluttered, and the change was rolled back soon after. ChatGPT Atlas, a stand alone browser, lasted less than a year before discontinuation on August 9. Operator, a browsing agent, was also folded into ChatGPT, while Sora shut down in March 2026 amid high costs and retention challenges.

Apple and Microsoft show slower paths. Apple unveiled Apple Intelligence in 2024 with an improved Siri able to understand context across apps. Repeated delays tied to engineering issues and bugs contributed to a 250 million dollar settlement over marketing of iPhone 16 AI capabilities. Siri AI reached the iOS 27 beta this past July.

DISCLOSURE

Researchers analyzed approximately 824K tweets alongside 266K comparison tweets with LIWC and an LLM-based intimacy classifier.

SAFETY

Kill Switch Fails Against Distributed AI Systems

Regulation debate focuses on shutdown power while models run across data centers and human oversight.

Calls for an AI kill switch misunderstand how advanced systems operate. The models run as distributed infrastructure across data centers and networks, not as a single machine that can be switched off. The metaphor hides technical complexity and human responsibility, making it a poor basis for national regulation.

Several efforts at AI regulation in 2026 have centered on the idea. Proposed congressional legislation includes the AI Kill Switch Act. California Gov. Gavin Newsom issued an executive order aimed at reining in AI. President Donald Trump has opposed the idea as unnecessary regulation. To date there has been no national regulation that includes a kill switch.

Research engineer Dylan Baker at the nonprofit Distributed AI Research Institute has explained the central technical problem. Current advanced AI models involve a complex infrastructure of distributed technologies. That infrastructure is beyond the control of a single kill switch. Advanced models that companies like OpenAI and Anthropic are concerned about releasing rely on multiple computers and data centers that are spread out nationally.

> **“the black box of AI cultivates useful confusion about what can and cannot be known”**
>
> ERYK SALVAGGIO, UNIVERSITY OF CAMBRIDGE

The critique extends to language. The kill switch belongs to the same rhetoric as the black box metaphor, which frames inner workings as confined and unintelligible. Both are what cognitive linguist George Lakoff and philosopher Mark Johnson call structural metaphors. Such metaphors use simple physical concepts like object, substance, up-down and in-out to clarify a complex idea. They also hide aspects of a concept by focusing attention on others. Here the focus on on or off hides the complexity of advanced models.

The second problem involves people.

WHAT THE PAPERS FOUND

Prediction changes the timing of decisions. A study in CARLA tests three driving strategies in risky merging scenarios: reactive driving, prediction-based longitudinal action, and prediction-based lateral action. The model combines knowledge graph embeddings with Bayesian inference. It anticipates a surrounding vehicle lane change approximately 3-4 seconds before occurrence. The premise is direct — predicting intentions lets an autonomous vehicle anticipate hazards and respond proactively rather than relying solely on reactive maneuvers.

Classroom and market studies test whether model assistance transfers. In an undergraduate proof-based mathematics course, 153 students received staggered access to an AI chatbot and homework-feedback interface. Early access lifted homework scores by 2.56 points, but exam performance did not differ significantly between groups. After all students received access, every additional 100 messages to the chatbot was associated with a 3.02-point lower score on the subsequent exam. In a longitudinal paper-trading study with 80 participants, no evaluated LLM reliably outperforms a simple recent-activity persistence baseline for predicting trade occurrence.

Mental health work measures disclosure and response. One study analyzes approximately 824K tweets across anxiety, depression, bipolar disorder, borderline personality disorder, and suicidal ideation or related experiences, alongside 266K comparison tweets. It uses Linguistic Inquiry and Word Count and an LLM-based intimacy classifier to characterize psycholinguistic expression around Elon Musk October 2022 acquisition of Twitter. A separate benchmark, VEILED-SH, provides 10,003 prompts spanning counselor chat, forum posts, platform search, and triage. It distinguishes intent, communicative function, and response behavior. Six automated profiles, individually calibrated, evaluate single-turn responses to indirect expressions of distress.

Efficiency and oversight shape deployment. DeepSeek-V4.1-Flash allocates 196 billion parameters to its Engram n-gram memory, motivating low-precision storage for pairs or triples of consecutive tokens. VERA, an open-source self-hosted dashboard deployed within a large European bank, organizes evidence under six principle groups and eighteen technical requirements, twelve of which current benchmarks can measure.